

BCD

Nese kalohet numri i bitave [9(8-bit)], mbledhim te dy anet me +6 (0110=>BCD Code).



Kodet BCD, xs-3

Press Esc to exit full screen

BCD $\rightarrow$ Binary Coded Decimal

2

0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	
1011	
1100	
1101	
1110	
1111	

+6

✓ ✓
27
+ 89

0110 ^x3
+5

1 + 1000
1011
+ 0110
0001
1
0001
1
0110
6

XS-3

Mbledhja:

Nese kemi bartje, mbledhim me +3 (0011).

Nese nuk kemi bartje, zbresim me -3(0011).

Zbritja:

Nese kemi huazim, zbresim me -3(0011).

Nese nuk kemi huazim, mbledhim me +3(0011).



Kodet BCD, XS-3

XS-3 Excess-3

0000	}	X
0001		
0010		
		0
0011		1
0100		2
0101		3
0110		4
0111		5
1000		6
1001		7
1010		8
1011		9
1100	}	X
1101		
1110		
1111		

$$\begin{array}{r}
 \begin{array}{l}
 \text{1111} \leftarrow \\
 0111 \quad 1000 \\
 + \quad 0101 \quad 1010 \\
 \hline
 1001 \quad 0010 \\
 - 0011 \quad + 0011 \\
 \hline
 1010 \quad 0101 \\
 7 \quad 2
 \end{array}
 \end{array}$$

Algjebra e BOOLE-it

AND OR NOT
DHE OSE JO
 $A*B$ $A+B$ $\bar{A}, (A')$
 AB

A	B	$A*B$	$A+B$	$\bar{A}$	$\bar{B}$	$A\bar{B}$	$\bar{A}\bar{B}$
0	0	0	0	1	1	0	1
0	1	0	1	1	0	0	0
1	0	0	1	0	1	1	1
1	1	1	1	0	0	0	0



Ligjet e BOOLE-it

$$1. A \bullet A = A$$

$$2. A + A = A$$

$$3. A \bullet \bar{A} = 0$$

$$4. A + \bar{A} = 1$$

$$\bar{\bullet} = +$$

$$5. A \bullet 0 = 0$$

$$6. A + 0 = A$$

$$\bar{+} = \bullet$$

$$7. A \bullet 1 = A$$

$$8. A + 1 = 1$$

$$9. \bar{\bar{A}} = A$$

$$10. \overline{A \bullet B} = \bar{A} + \bar{B} \Rightarrow \text{Ligji i De Morgan-it}$$

$$\overline{A + B} = \bar{A} \bullet \bar{B}$$

Activate Windows
Go to Settings to activate Windows.



$$A+B = B+A$$

$$A \bullet B = B \bullet A$$

$$A+(B+C) = (A+B)+C$$

$$A \bullet (B \bullet C) = (A \bullet B) \bullet C$$

$$A \bullet (B+C) = A \bullet B + A \bullet C$$

$$A+B \bullet C = (A+B) \bullet (A+C)$$

$$(A+B) \bullet (A+C) = A \bullet A + B \bullet A + A \bullet C + B \bullet C = A+B \bullet A + A \bullet C + B \bullet C = A (1+B+C) + BC = A1 + BC = A+B \bullet C.$$



Algebra e Boole-it $m \rightarrow \text{minterm}$

s_1	s_2	s_3	
A	B	C	F
0	0	0	0 M_1
0	0	1	1 m_1
0	1	0	0 M_2
0	1	1	1 m_2
0	1	1	1 m_3
1	0	0	1 m_4
1	0	1	0 M_3
1	1	0	0 M_4
1	1	1	1

$$F = m_1 + m_2 + m_3 + \dots$$

$M \rightarrow \text{maxterm}$

$$\bar{F} = M_1 \cdot M_2 \cdot M_3 \dots$$

$$m_1 = \bar{A}\bar{B}C \quad m_2 = \bar{A}BC$$

$$m_3 = A\bar{B}\bar{C} \quad m_4 = A\bar{B}C$$

$$F = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C$$



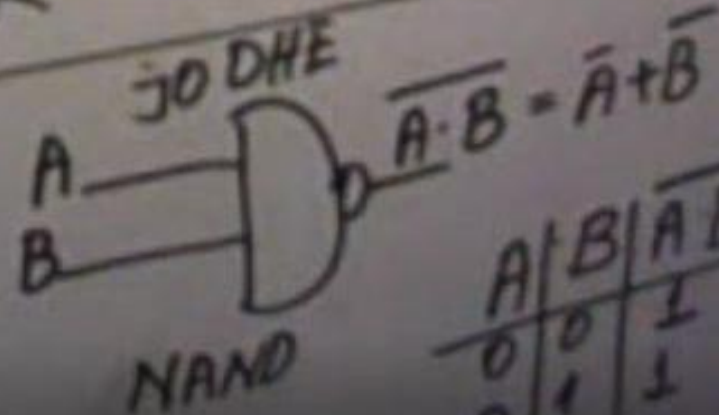
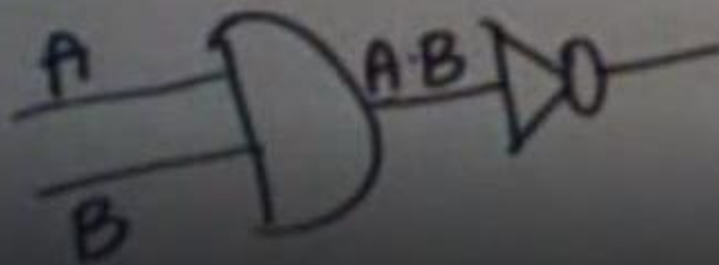
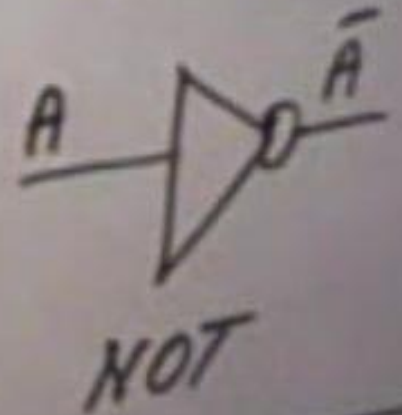
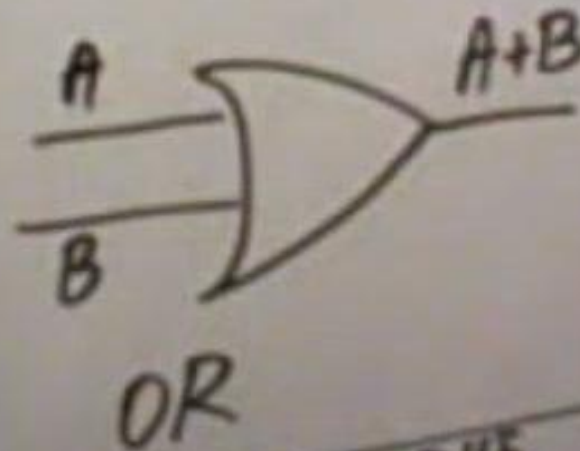
Target Digital Kombinasie

$$\overline{\overline{A+B}} = \overline{\overline{A}} \cdot \overline{\overline{B}}$$

DHE

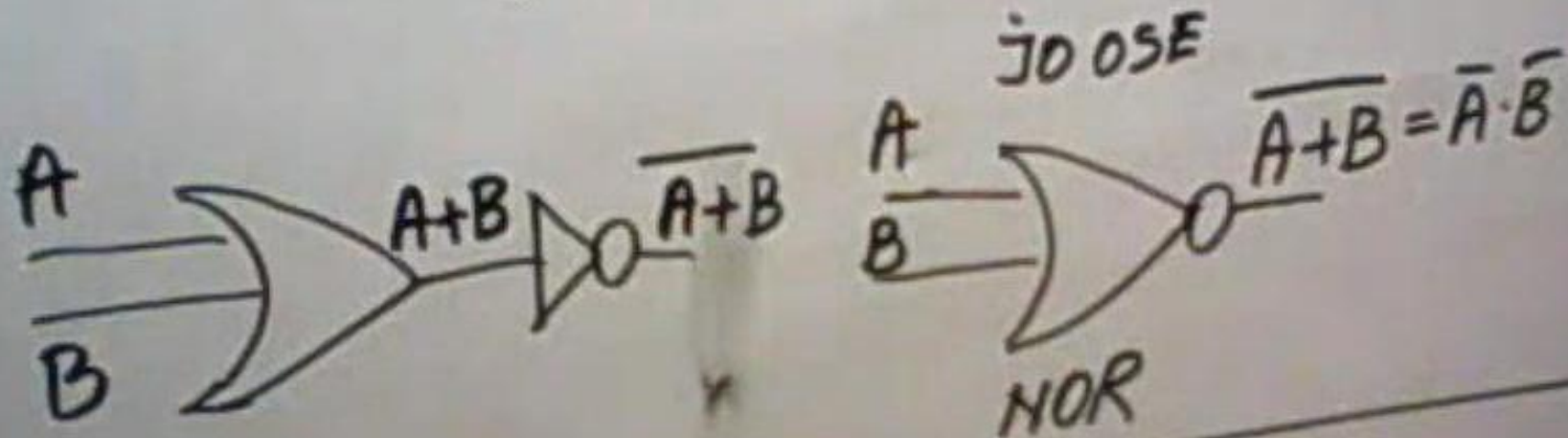
OSE

JO

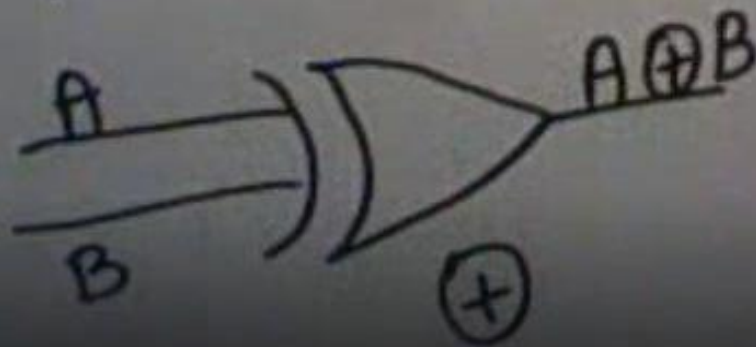


A	B	$\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

Target Digital Kombinaese



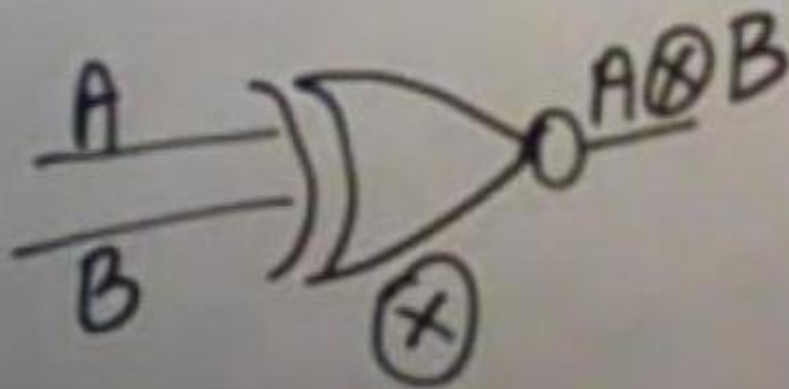
XOR $\rightarrow$ Exclusive OR



A	B	$A \oplus B$	$A \oplus B$
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

Target Digital Kombi

XNOR $\rightarrow$ Exclusive NOT OR



OR - OSE

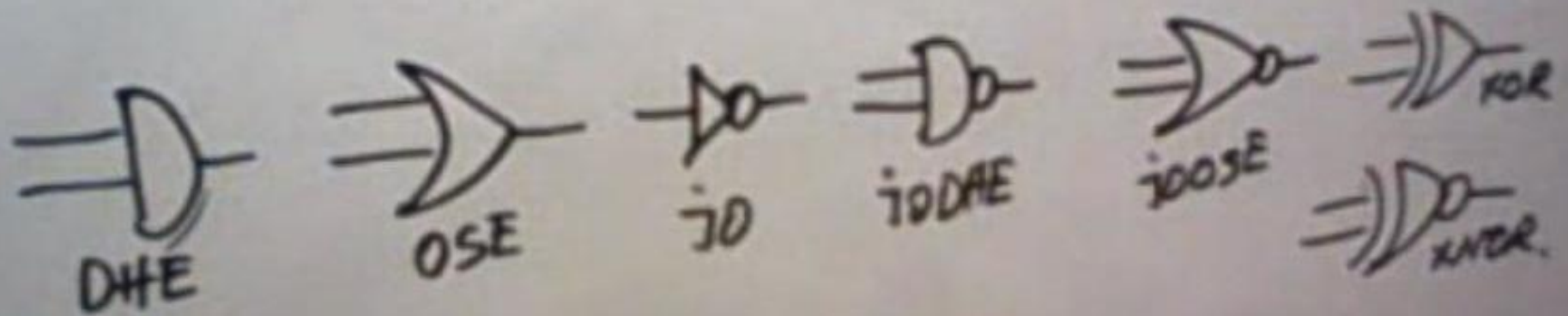
A	B	$A \oplus B$
0	0	1 m_1
0	1	0
1	0	0
1	1	1 m_2

1	0
0	1

$$A \oplus B = \bar{A} \cdot \bar{B} + A \cdot B$$

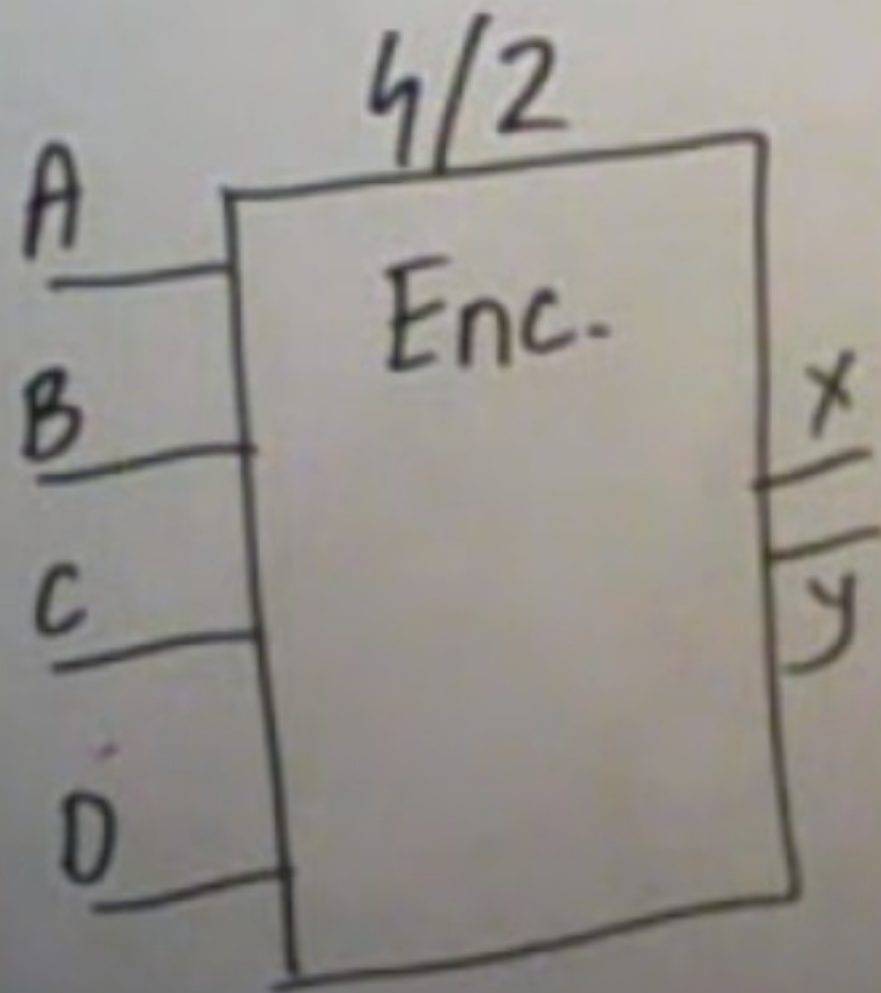


Qarget Digjital Kombinaese

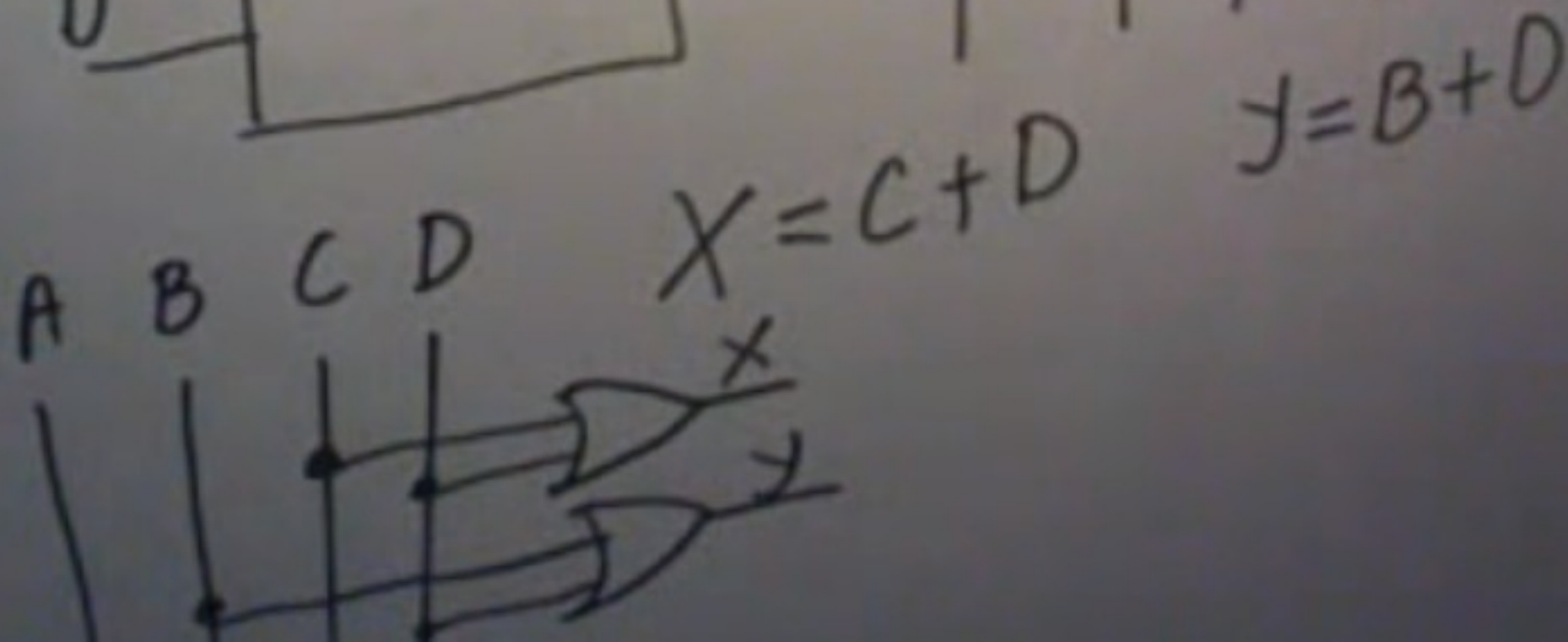


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Koduesit (Encoders)

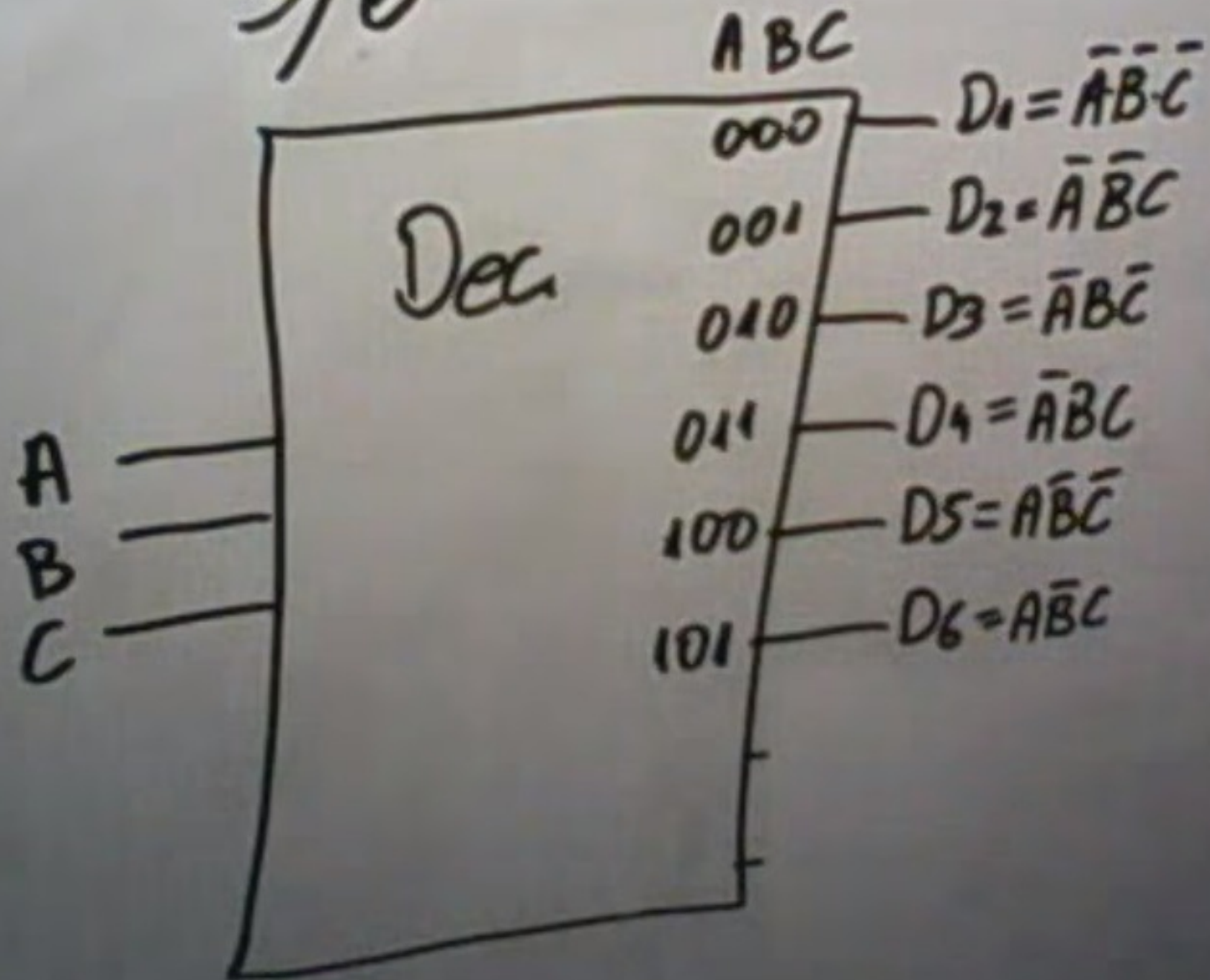


A	B	C	D	x	y
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

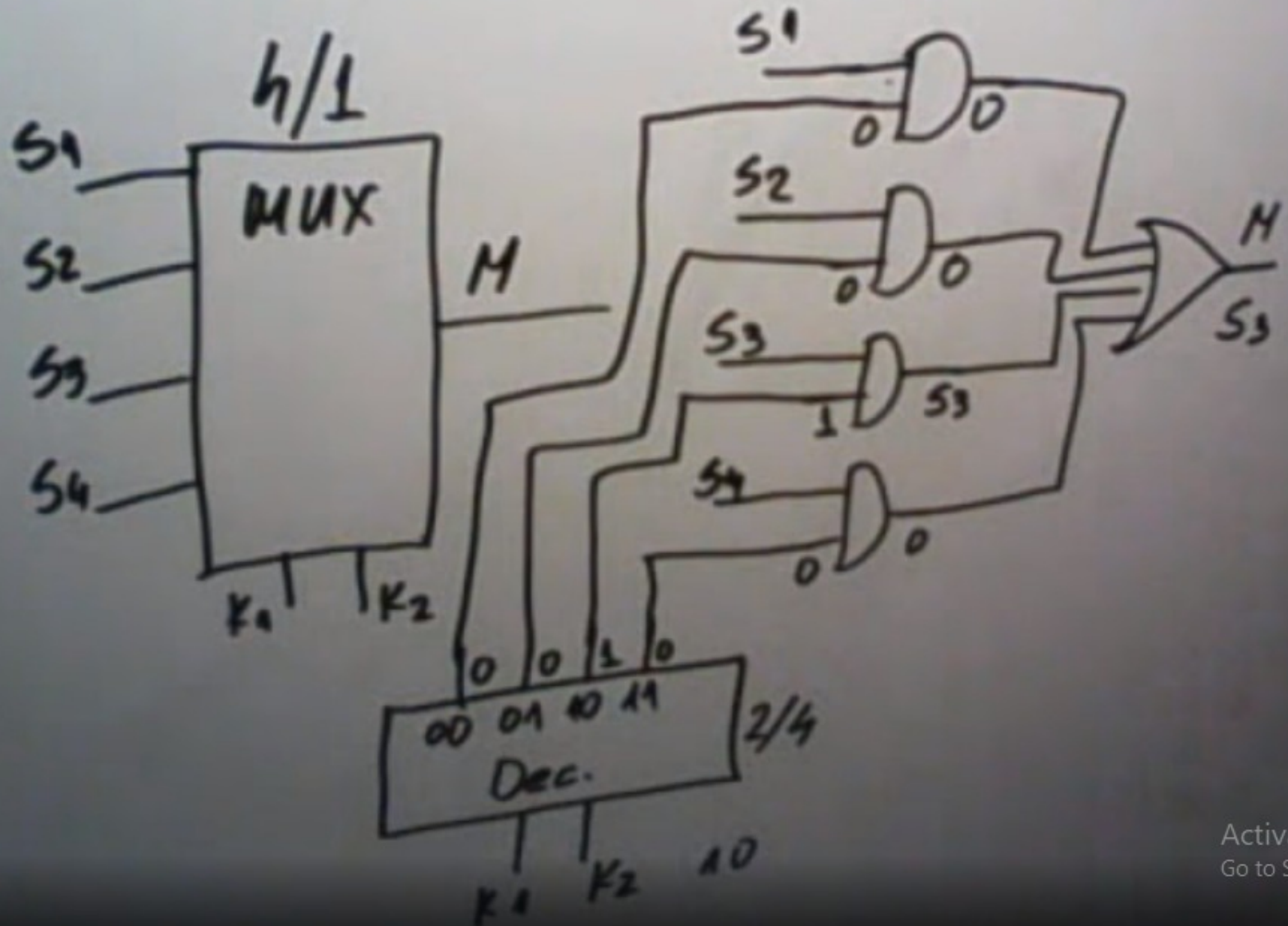


Dekoduesit (Decoders)

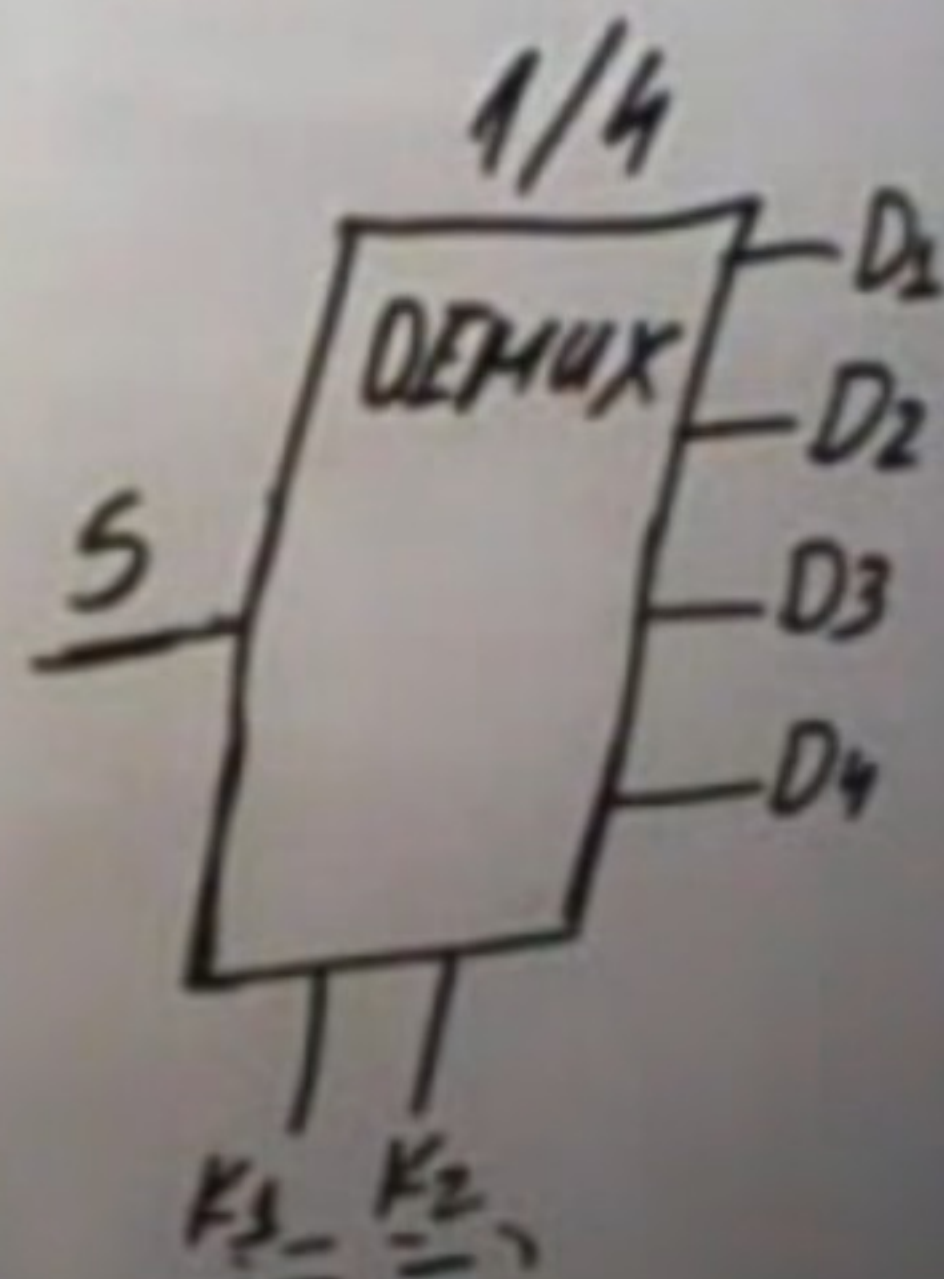
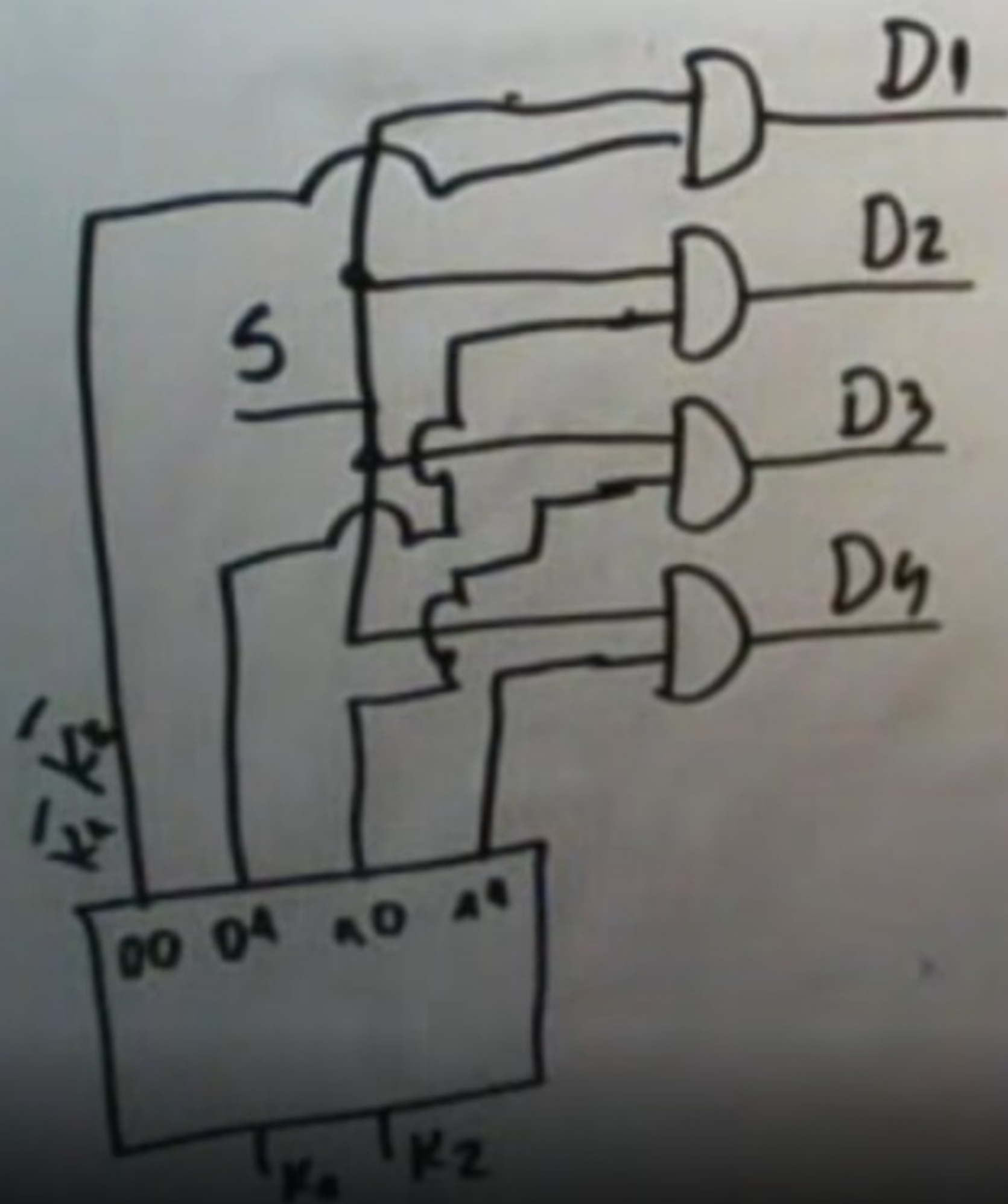
3/6



Multiplexer (Mux), Demultiplexer (Demux)



Multiplexer (Mux), Demultiplexer (DEMUX)



$$D_1 = S \cdot K_1 \cdot K_2$$

$$D_2 = S \cdot \bar{K}_1 \cdot K_2$$

$$D_3 = S \cdot K_1 \cdot \bar{K}_2$$

$$D_4 = S \cdot \bar{K}_1 \cdot \bar{K}_2$$

Multiplexeret (Mux)

